
GOAL
Track how the proportion and activation of reward-encoding neurons change across training days, by brain region and cohort (supervised vs unsupervised). Prediction: both rise in anterior HVAs, supervised only.
SHARED INPUTS
Neural data — up to ~80k neurons, SVD(400-PC)-compressed; reconstruct a large sample per region (V1 · medial · lateral · anterior)
Behavior — sound-cue position (≈ reward zone, present with OR without water), trial structure, corridor position, running
PRIMARY — d′(late vs early cue), across days
SAMPLE NEURONS per region per mouse (750 per region; capped at what a region has)
REWARD-ZONE ACTIVITY per rewarded-corridor (leaf1) trial, mean activity in the reward zone (5–40 dm)
SPLIT TRIALS by cue position early-cue vs late-cue (cue tracks reward; exists in both cohorts, so the comparison is fair)
CROSS-VALIDATED d′ (per neuron) two interleaved trial folds; d′ = 2(μ_late − μ_early)/(σ_late + σ_early)
CLASSIFY (flag) reward-encoding if |d′| ≥ 0.3 in both folds, same sign — noise rarely survives both
ACTIVATION (strength) mean |d′| over folds, for all neurons (unbiased)
AGGREGATE — mouse is the replicate per mouse: % reward-encoding (how many) and mean d′ (how strongly); mean ± s.e.m. across mice, across days × region × cohort
RIGOR / CONTROLS
cross-validation: effect must reproduce in both folds · shuffle null on late/early labels · cue-based reward event (cross-cohort fair) · mouse as replicate (neurons are not independent) · skip + report sessions with no cue variation or too few trials
SHUFFLE FLOOR scramble late/early labels, rerun the same cross-validated test → chance % expected from noise
INTERPRETATION Above the chance floor, rising before→after learning, concentrated in anterior HVAs, supervised only ⇒ supports the reward-prediction account. Unsupervised should stay near the floor.